

6(a)	force <u>per</u> unit positive charge	B1
6(b)(i)	$E_K = \frac{1}{2}mv^2$	C1
	$2.4 \times 10^{-16} = \frac{1}{2} \times 1.7 \times 10^{-27} \times v^2$	A1
	$v = 5.3 \times 10^5 \text{ ms}^{-1}$	
6(b)(ii)	work done = $2.4 \times 10^{-16} \text{ J}$	A1
6(b)(iii)	$W = Fs$	C1
	$F = 2.4 \times 10^{-16} / 15 \times 10^{-3}$	A1
	$= 1.6 \times 10^{-14} \text{ N}$	
6(b)(iv)	$V = Fd / Q$ or $V = W / Q$ or $E = V/d$ and $E = F / Q$	C1
	$V = (1.6 \times 10^{-14} \times 15 \times 10^{-3}) / 1.6 \times 10^{-19}$ or $2.4 \times 10^{-16} / 1.6 \times 10^{-19}$	C1
	$= 1500 \text{ V}$	A1
6(b)(v)	straight line with positive gradient starting at the origin and going as far as $x = 15 \text{ mm}$	B1